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CHAPTER

Introduction[*]

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1.1 The historical context

During the period that covers, roughly, the last 250 years, humankind has lived and experienced three *transforming* revolutions, which have been powered by technology and science. The *first industrial* revolution was based on the use of water and steam and its origins are traced to the end of the 18th century, when the first organized factories appeared in England. The *second industrial* revolution was powered by the use of electricity and mass production, and its "birth" is traced back to around the turn of the 20th century. The *third industrial* revolution was fueled by the use of electronics, information technology, and the adoption of automation in production. Its origins coincide with the end of the Second World War.

Although difficult for humans, including historians, to put a stamp on the age in which they themselves live, more and more people are claiming that the *fourth industrial* revolution has already started

[*] This book has a companion website hosting complementary materials. Visit this URL to access it: https://www.elsevier.com/books-and-journals/book-companion/9780443292385.

Machine Learning. https://doi.org/10.1016/B978-0-44-329238-5.00007-X
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and is fast transforming everything that we know and learned and live with so far. The fourth industrial revolution builds upon the third one and is powered by the *fusion* of a number of technologies, e.g., computers and communications (internet), and it is characterized by the convergence of the *physical*, *digital*, and *biological* spheres.

The terms artificial intelligence (AI) and machine learning are used and spread more and more to denote the type of automation technology that is used in the production (industry), in the distribution of goods (commerce), in the service sector, and in our economic transactions (e.g., banking). Moreover, these technologies affect and shape the way we socialize and interact as humans via social networks, and the way we entertain ourselves, involving games and cultural products such as music and movies.

A distinct qualitative difference of the fourth, compared to the previous industrial revolutions, is that, before, it was the manual skills of humans that were gradually replaced by "machines." In the one that we are currently experiencing, mental skills are also replaced by "machines." We now have automatic answering software that runs on computers, fewer people are serving us in banks, and many jobs in the service sector have been taken over by computers and related software platforms. Soon, we are going to have cars without drivers and drones for deliveries. At the same time, new jobs, needs, and opportunities appear and are created. The labor market is fast changing and new competences and skills are and will be required in the future (see, e.g., [22,23]).

At the center of this historical happening, as one of the key enabling technologies, lies a discipline that deals with data and whose goal is to extract information and related knowledge that is hidden in it, in order to make predictions and, subsequently, take decisions. That is, the goal of this discipline is to *learn* from data. This is analogous to what humans do in order to reach decisions. Learning through the senses, personal experience, and the knowledge that propagates from generation to generation is at the heart of human intelligence. Also, at the center of any scientific field lies the development of models (often called theories) in order to explain the available experimental evidence. In other words, *data* comprise a major source of *learning*.

1.2 Artificial intelligence and machine learning

The title of the book refers to *machine learning*, although the term *artificial intelligence* is used more and more, especially in the media but also by some experts, to refer to any type of algorithms and methods that perform tasks that traditionally required human intelligence. Being aware that definitions of terms can never be exact and there is always some "vagueness" around their respective meanings, I will still attempt to clarify what I mean by machine learning and in which aspects this term means something different from AI. No doubt, there may be different views on this.

Although the term machine learning was popularized fairly recently, as a scientific field it is an old one, whose roots go back to statistics, computer science, information theory, signal processing, and automatic control. Examples of some related names from the past are statistical learning, pattern recognition, adaptive signal processing, system identification, image analysis, and speech recognition. What all these disciplines have in common is that they process data, develop models that are *data-adaptive*, and subsequently make predictions that can lead to decisions. Most of the basic theories and algorithmic tools that are used today had already been developed and known before the dawn of this century. With a "small" yet important difference: the available data, as well as the computer power prior to 2000, were not enough to use some of the more elaborate and complex models that had been

developed. The terrain started changing after 2000, in particular around 2010. Large data sets were gradually created and the computer power became affordable to allow the use of more complex models. In turn, more and more applications adopted such algorithmic techniques. "Learning from data" became the new trend and the term machine learning prevailed as an umbrella for such techniques.

Moreover, the big difference was made with the use and "rediscovery" of what is today known as *deep neural networks*. These models offered impressive predictive accuracies that had never been achieved by previous models. In turn, these successes paved the way for the adoption of such models in a wide range of applications and also ignited intense research, and new versions and models have been proposed. These days, another term that is catching up is "data science," indicating the emphasis on how one can develop robust machine learning and computational techniques that deal efficiently with large-scale data.

However, the main rationale, which runs through the spine of all the methods that come under the machine learning umbrella, remains the same and it has been around for many decades. The main concept is to estimate a set of parameters that describe the model, using the available data and, in the sequel, to make *predictions* based on *low-level information and signals*. One may easily argue that there is not much *intelligence* built in such approaches. No doubt, deep neural networks involve much more "intelligence" than their predecessors. They have the potential to *optimize* the *representation* of their low-level input information to the computer.

The term "representation" refers to the way in which related information that is hidden in the input data is quantified/coded so that it can be subsequently processed by a computer. In the more technical jargon, each piece of such information is known as a feature (see also Section 1.5.1). As discussed in detail in Chapters 18 and 19, where neural networks (NNs) are defined and presented in detail, what makes these models distinctly different from other data learning methods is their *multilayer* structure. This allows for the "building" up of a *hierarchy* of representations of the input information at various abstraction levels. Every layer builds upon the previous one, and the higher in hierarchy, the more abstract the obtained representation is. This structure offers to neural networks a significant performance advantage over alternative models, which restrict themselves to a single representation layer. Furthermore, this single-level representation was rather hand-crafted and designed by the users, in contrast to the deep networks that "learn" the representation layers from the input data via the use of optimality criteria.

Yet, in spite of the previously stated successes, I share the view that we are still very far from what an intelligent machine should be. For example, once trained (estimating the parameters) on one data set, which has been developed for a specific task, it is not easy for such models to *generalize* to other tasks. Although, as we are going to see in Chapters 18 and 19, advances in this direction have been made, we are still very far from what human intelligence can achieve. When a child sees one cat, the child readily recognizes another one, even if this other cat has a different color or if it turns around. Current machine learning systems need thousands of images with cats in order to be trained to "recognize" one in an image. If a human learns to ride a bike, it is very easy to transfer this knowledge and learn to ride a motorbike or even to drive a car. Humans can easily transfer knowledge from one task to another, without forgetting the previous one. In contrast, current machine learning systems lack such a generalization power and tend to forget the previous task once they are trained to learn a new one. This is also an open field of research, where advances have also been reported.

Furthermore, machine learning systems that employ deep networks can even achieve superhuman prediction accuracies on data similar to those with which they have been trained. This is a significant

achievement, not to be underestimated, since such techniques can efficiently be used for dedicated jobs, for example, to recognize faces, to recognize the presence of various objects in photographs, and also to annotate images and produce text that is related to the content of the image. They can recognize speech, translate text from one language to another, detect which music piece is currently playing in the bar, and determine whether the piece belongs to the jazz or to the rock musical genre. At the same time, they can be fooled by carefully constructed examples, known as adversarial examples, in a way that no human would be fooled to produce a wrong prediction (see Chapter 18).

Concerning AI, the term "artificial intelligence" was first coined by John McCarthy in 1956 when he organized the first dedicated conference (see, e.g., [20] for a short history). The concept at that time, which still remains a goal, was whether one can build an intelligent machine, realized on software and hardware, that can possess *human-like* intelligence. In contrast to the field of machine learning, the concept for AI was not to focus on low-level information processing with emphasis on predictions, but on the high-level *cognitive* capabilities of humans to *reason* and *think*. No doubt, we are still very far from this original goal. Predictions are, indeed, part of intelligence. Yet, intelligence is much more than that. Predictions are associated with what we call *inductive* reasoning. Yet what really differentiates human from the animals intelligence is the power of the human mind to form *concepts* and *create conjectures* for explaining data and more general the world in which we live. *Explanations* comprise a high-level facet of our intelligence and constitute the basis for scientific theories and the creation of our civilization. They are assertions concerning the "why"'s and the "how"'s related to a task (e.g., [5,6,11]).

To talk about AI, at least as it was conceived by pioneers such as Alan Turing [16], systems should have built-in capabilities for *reasoning* and giving *meaning*, e.g., in language processing, to be able to infer *causality*, to model efficient representations of *uncertainty*, and, also, to pursue long-term goals [8]. Possibly, toward achieving these challenging goals, we may have to understand and implement notions from the theory of mind, and also build machines that implement self-awareness. The former psychological term refers to the understanding that others have their own beliefs and intentions that justify their decisions. The latter refers to what we call *consciousness*. As a last point, recall that human intelligence is closely related to feelings and emotions. As a matter of fact, the latter seem to play an important part in the *creative* mental power of humans (e.g., [3,4,17]). Thus, in this more theoretical perspective AI still remains a *vision for the future*.

The previous discussion should not be taken as an attempt to get involved with philosophical theories concerning the nature of human intelligence and AI. These topics comprise a field in itself, for more than 60 years, which is much beyond the scope of this book. My aim was to make the newcomer in the field aware of some views and concerns that are currently being discussed.

In the more practical front, for the early years, the term AI was used to refer to techniques built around knowledge-based systems that sought to hard-code knowledge in terms of formal languages, e.g., [13]. Computer "reasoning" was implemented via a set of logical inference rules. In spite of the early successes, such methods seem to have reached a limit (see, e.g., [7]). It was the alternative path of machine learning, via learning from data, that gave a real push into the field. These days, the term AI is used as an umbrella to cover all methods and algorithmic approaches that are related to the machine intelligence discipline, with machine learning and knowledge-based techniques being parts of it.

1.3 Algorithms can learn what is hidden in the data

It has already been emphasized that data lie at the heart of machine learning systems. Data are the *beginning*. It is the information hidden in the data, in the form of underlying regularities, correlations, or structure, which a machine learning system tries to "learn." Thus, irrespective of how intelligent a software algorithm is designed to be, it cannot learn more than what the data which it has been trained on allow.

Collecting the data and building the data set on which an "intelligent" system is going to be trained is highly critical. Building data sets that address human needs and developing systems that are going to make decisions on issues, where humans and their lives are involved, requires special attention, and above all, responsibility. This is not an easy issue and good intentions are not enough. We are all products of the societies in which we live, which means that our beliefs, to a large extent, are formed by the prevailing social stereotypes concerning, e.g., gender, racial, ethnic, religious, cultural, class-related, and political views. Most importantly, most of these beliefs take place and exist at a subconscious level. Thus, sampling "typical" cases to form data sets may have a strong flavor of *subjectivity* and introduce biases. A system trained on such data can affect lives, and it may take time for this to be found out. Furthermore, our world is fast changing and these changes should continuously be reflected in the systems that make decisions on our behalf. Outsourcing our lives to computers should be done cautiously and above all in an ethical framework, which is much wider and general than the set of the existing legal rules.

Of course, although this puts a burden on the shoulders of the individuals, governments, and companies that develop data sets and "intelligent" systems, it cannot be left to their good will. On the one hand, a specialized legal framework that guides the *design*, *implementation*, and *use* of such platforms and systems is required to protect our ethical standards and social values. Of course, this does not concern only the data that are collected but also the overall system that is built. Any system that replaces humans should be (a) transparent, (b) fair, and (c) accurate. Not that the humans act, necessarily, according to what the previous three terms mean. However, humans can, also, reason and discuss, we have feelings and emotions, and we do not just perform predictions.

On the other hand, this may be the time when we can develop and build more "objective" systems, that is, to go beyond human subjectivity. However, such "objectivity" should be based on science, rules, criteria, and principles, which are not yet here. As Michael Jordan [8] puts it, the development of such systems will require perspectives from the *social sciences* and *humanities*. Currently, such systems are built following an ad hoc rather than a principled way. Karl Popper [12], one of the most influential philosophers of science, stressed that all knowledge creation is theory-laden. Observations are never free of an underlying theory or explanation. Even if one believes that the process begins with observations, the act of observing requires a *point of view* (see, also, [1,6]).

If I take the liberty to make a bit of a science fiction (something trendy these days), when AI, in the context of its original conception, is realized, then data sampling and creation of data sets for training could be taken care of by specialized algorithms. Maybe such algorithms will be based on scientific principles that in the meantime will have been developed. After all, this may be the time of dawn for the emergence of a new scientific/engineering field that integrates in a principle way data-focused disciplines. To this end, another statement of Karl Popper may have to be implemented, i.e., that of *falsification*, yet in a slightly abused interpretation. An emphasis on building intelligent systems should be directed on criticism and experimentations for finding evidence that *refutes* the principles,

which were employed for their development. Systems can only be used if they survive the falsification test.

1.4 Typical applications of machine learning

It is hard to find a discipline in which machine learning techniques for "learning" from data have not been applied. Yet, there are some areas, which can be considered as typical applications, maybe due to their economic and social impact. Examples of such applications are summarized below.

Speech recognition

Speech is the primary means of communication among humans. Language and speech comprise major attributes that differentiate humans from animals. Speech recognition has been one of the main research topics whose roots date back to the early 1960s. The goal of speech recognition is to develop methods and algorithms that enable the recognition and the subsequent representation in a computer of spoken language. This is an interdisciplinary field involving signal processing, machine learning, linguistics, and computer science.

Examples of speech recognition tasks and related systems that have been developed over the years range from the simplest isolated word recognition, where the speaker has to wait between utterances, to the more advanced continuous speech recognizers. In the latter, the user can speak almost naturally, and concurrently the computer can determine the content. Speaker recognition is another topic, where the system can identify the speaker. Such systems are used, for example, for security purposes.

Speech recognition embraces a wide spectrum of applications. Some typical cases where speech recognizers have been used include automatic call processing in telephone networks, query-based information systems, data entry, voice dictation, robotics, as well as assistive technologies for people with special needs, e.g., blind people.

Computer vision

This is a discipline that has been inspired by the human visual system. Typical tasks that are addressed within the computer vision community include the automatic extraction of edges from images, representation of objects as compositions of smaller structures, object detection and recognition, optical flow, motion estimation, inference of shape from various cues, such as shading and texture, and three-dimensional reconstruction of scenes from multiple images. Image morphing, that is, changing one image to another through a seamless transition, and image stitching, i.e., creating a panoramic image from a number of images, are also topics in the computer vision research. Image generation, e.g., text-to-image generation, is another popular trend, which is treated in Chapter 19. Over the recent years, there is more and more interaction between the field of computer vision and that of graphics.

Multimodal data

Both speech recognition and computer vision process information that originates from single *modalities*. However, humans perceive the natural world in a *multimodal way*, via their multiple senses, e.g.,

vision, hearing, and touch. There is complementary information in each one of the involved modalities that the human brain exploits in order to understand and perceive the surrounding world.

Inspired by that, *multimedia* or multimodal understanding, via *cross-media integration*, has given birth to a related field whose goal is to improve the performance in the various scientific tasks that arise in problems that deal with multiple modalities. An example of modality blending is to combine together image/video, speech/audio, and text. A related summary that also touches issues concerning the mental processes of human *sensation*, *perception*, and *cognition* is presented in [9].

Natural language processing

This is the discipline that studies the processing of a language using computers. An example of a natural language processing (NLP) task is that of SPAM detection. Currently, the NLP field is an area of intense research with typical topics being the development of automatic translation algorithms and software, sentiment analysis, text summarization, authorship identification, text generation. Speech recognition has a strong affinity with NLP and, strictly speaking, could be considered as a special subtopic of it. The field of NLP has recently seen impressive advances in the context of the so-called large language models (LLMs), which are treated in Chapter 19.

Robotics

Robots are used to perform tasks in the manufacturing industry, e.g., in an assembly line for car production, or by space agencies to move objects in the space. More recently, the so-called social robots are built to interact with people in their social environment. For example, social robots are used to benefit hospitalized children [10].

Robots have been used in situations that are difficult or dangerous for humans, such as bomb detonation and work in difficult and hazardous environments, e.g., places of high heat, deep oceans, and areas of high radiation. Robots have also been developed for teaching.

Robotics is an interdisciplinary field that, besides machine learning, includes disciplines such as mechanical engineering, electronic engineering, computer science, computer vision, and speech recognition.

Autonomous cars

An autonomous or *self-driving* car is a vehicle that can move around with no or little human intervention. Most of us have used self-driving trains in airports. However, these operate in a very well-controlled environment. Autonomous cars are designed to operate in the city streets and in motorways. This field is also of interdisciplinary nature, where areas such as radar, lidar, computer vision, automatic control, sensor networks, and machine learning meet together. It is anticipated that the use of self-driving cars will reduce the number of accidents, since, statistically, most of the accidents occur because of human errors, due to alcohol, high speed, stress, fatigue, etc.

There are various levels of automation that one can implement. At level 0, which is the category in which most of the cars currently operate, the driver has the control and the automated built-in system may issue warnings. The higher the level, the more autonomy is present. For example, at level 4, the driver would be first notified whether conditions are safe, and then the driver can decide to switch

the vehicle into the autonomous driving mode. At the highest level, level 5, the autonomous driving requires absolutely no human intervention [21].

Besides the aforementioned examples of notable machine learning applications, machine learning has been applied in a wide range of other areas, such as healthcare, bioinformatics, business, finance, education, law, and manufacturing.

Challenges for the future

In spite of the impressive advances that have been achieved in machine learning, there are a number of challenges for the foreseeable future, besides the long-term ones that were mentioned before, while presenting AI. In the Berkeley report [18], the following list of challenges is summarized:

- Designing systems that learn *continually* by interacting with a *dynamic environment*, while making decisions that are *timely*, *robust*, and *secure*.
- Designing systems that enable personalized applications and services, yet do not compromise users' *privacy and security*.
- Designing systems that can train on data sets owned by different organizations without compromising their *confidentiality*, and in the process provide AI capabilities that span the boundaries of potentially competing organizations.
- Developing domain-specific architectures and software systems to address the performance needs of future applications, including custom chips, edge-cloud systems to efficiently process data at the edge, and techniques for abstracting and sampling data.

Besides the above more technology-oriented challenges, important social challenges do exist. The new technologies are influencing our daily lives more and more. In principle, they offer the potential, much more than ever before, to manipulate and shape beliefs, views, interests, entertainment, customs, and culture, independent of the societies. Moreover, they offer the potential for accessing personal data that in the sequel can be exploited for various reasons, such as economic, political, or other malicious purposes. As M. Schaake, a member of the European Parliament, puts it, "When algorithms affect human rights, public values or public decision-making, we need oversight and transparency." However, what was said before should not mobilize technophobic reactions. On the contrary, human civilization has advanced because of leaps in science and technology. All that is needed is social sensitivity, awareness of the possible dangers, and a related legal "shielding."

Putting it in simple words, as Henri Bergson said [2], history is not deterministic. History is a *creative* evolution.

1.5 Machine learning: major directions

As has already been stated before, machine learning is the scientific field whose goal is to develop methods and algorithms that "learn" from data; that is, to extract information that "resides" in the data, which can subsequently be used by the computer to perform a task. To this end, the starting point is an available data set. Depending on the type of information that one needs to acquire, in the context of a specific task, different types of machine learning have been developed. They are described below.

1.5.1 Supervised learning

Supervised learning refers to the type of machine learning where all the available data have been *labeled*. In other words, data are represented in pairs of *observations*, e.g., $(y_n, \boldsymbol{x}_n)$, $n = 1, 2, \ldots, N$, where each $\boldsymbol{x}_n$ is a vector or, in general, a set of variables. The variables in $\boldsymbol{x}_n$ are called the input variables, also known as the *independent* variables or *features*, and the respective vector is known as the *feature vector*. The variables y_n are known as the *output* or *dependent* or *target* or *label* variables. In many cases, y_n can also be a vector. The goal of learning is to obtain/estimate a functional mapping to predict the value of the output, given the values of the respective input variables. Two "pillars" of supervised learning are the classification and the regression tasks.

Classification

The goal in classification is to assign a *pattern* to one of a set of possible classes, whose number is considered to be known. For example, in X-ray mammography, we are given an image where a region indicates the existence of a tumor. The goal of a computer-aided diagnosis system is to predict whether this tumor corresponds to the *benign* or the *malignant* class. Optical character recognition (OCR) systems are also built around a classification system, in which the image corresponding to each letter of the alphabet has to be recognized and assigned to one of the 26 (for the Latin alphabet) classes; see Example 18.3 for a related case study. Another example is the prediction of the authorship of a given text. Given a text written by an unknown author, the goal of a classification system is to predict the author among a number of authors (classes); this application is treated in Section 11.15. The receiver in a digital communications system can also be viewed as a classification system. Upon receiving the transmitted data, which have been contaminated by noise and also by other transformations imposed by the transmission channel (Chapter 4), the receiver has to reach a decision on the value of the originally transmitted symbols. For example, in a binary transmitted sequence, the original symbols belong either to the $+1$ or to the -1 class. This task is known as *channel equalization*.

The first step in designing any machine learning task is to decide how to represent each pattern in the computer. This is achieved during the preprocessing stage; one has to "encode" related information that resides in the raw data (e.g., image pixels or strings of words) in an efficient and information-rich way. This is usually done by transforming the raw data into a new space and representing each pattern by a vector, $\boldsymbol{x} \in \mathbb{R}^l$. This comprises the feature vector and its l elements the corresponding feature values. In this way, each pattern becomes a single point in an l-dimensional space, known as the *feature space* or the *input space*. We refer to this transformation of the raw data as the *feature generation* or *feature extraction* stage. One starts with generating some large value, K, of possible features and eventually selects the l most informative ones via an optimizing procedure known as the *feature selection* stage. As we will see in Section 18.12, in the context of convolutional neural networks, the previous two stages are merged together and the features are obtained and optimized in a combined way, together with the estimation of the functional mapping that performs the prediction, as it was mentioned before.

Having decided upon the input space in which the data are represented, one has to train a classifier, that is, a predictor. This is achieved by first selecting a set of N data points/samples/examples, whose class is known, and this comprises the *training set*. This is the set of *observation* pairs, $(y_n, \boldsymbol{x}_n)$, $n = 1, 2, \ldots, N$, where y_n is the (output) variable denoting the class in which $\boldsymbol{x}_n$ belongs, and it is known as the corresponding *class label*; the class labels take values over a *discrete* set, e.g., $\{1, 2, \ldots, M\}$,

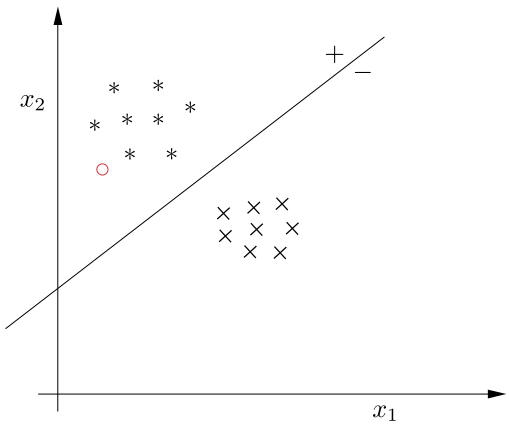


FIGURE 1.1

The classifier (linear in this simple case) has been designed in order to separate the training data into two classes. The graph (straight line) of the linear function, $f(\boldsymbol{x}) = 0$, has on its positive side the points coming from one class and on its negative side those of the other. The "red" point, whose class is unknown, is classified in the same class as the "star" points, since it lies on the positive side of the line.

for an M-class classification task. For example, for a two-class classification task, $y_n \in \{-1, +1\}$ or $y_n \in \{0, +1\}$. To keep our discussion simple, let us focus on the two-class case. Based on the training data, one then designs a function, f, which is used to predict the output label, given the input feature vector, $\boldsymbol{x}$. In general, we may need to design a set of such functions.

Once the function, f, has been designed, the system is ready to make predictions. Given a pattern whose class is unknown, we obtain the corresponding feature vector, $\boldsymbol{x}$, from the raw data. Depending on the value of $f(\boldsymbol{x})$, the pattern is classified in one of the two classes. For example, if the labels take the values ± 1, then the predicted label is obtained as $\hat{y} = \text{sgn}\{f(\boldsymbol{x})\}$. This operation defines the *classifier*. If the function f is linear (nonlinear) we say that the respective classification task is linear (nonlinear) or, in a slight abuse of terminology, that the classifier is a linear (nonlinear) one.

Fig. 1.1 illustrates the classification task. Initially, we are given the set of points, each one representing a pattern in the two-dimensional space (two features used, x_1, x_2). Stars belong to one class, and the crosses to the other, in a two-class classification task. These are the training points, which are used to obtain a classifier. For our very simple case, this is achieved via a linear function,

$$f(\boldsymbol{x}) = \theta_1 x_1 + \theta_2 x_2 + \theta_0, \tag{1.1}$$

whose graph, for all the points such that $f(\boldsymbol{x}) = 0$, is the straight line shown in the figure. The values of the parameters $\theta_1, \theta_2, \theta_0$ are obtained via an estimation method based on the training set. This phase, where a classifier is *estimated*, is also known as the *training* or *learning* phase.

Once a classifier has been "learned," we are ready to perform predictions, that is, to predict the class label of a pattern $\boldsymbol{x}$. For example, we are given the point denoted by the red circle, whose class is unknown to us. According to the classification system that has been designed, this belongs to the same class as the points denoted by stars, which all belong to, say, class $+1$. Indeed, every point on one side

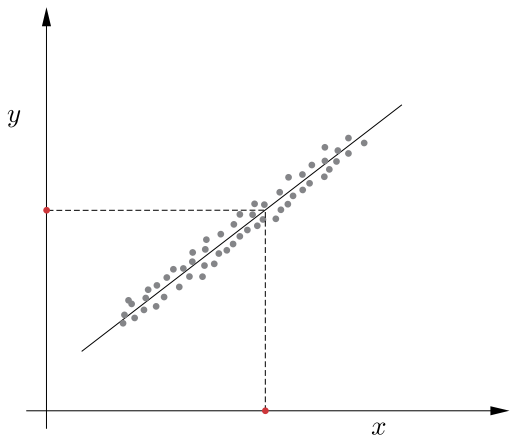


FIGURE 1.2

In a regression task, a function (linear in this case) f is designed such that its graph fits the available training data set. Given a new (red) point, x, the prediction of the respective output (red) value is computed as $\hat{y} = f(x)$.

of the straight line will give a positive value, $f(\boldsymbol{x}) > 0$, and all the points on its other side will give a negative value, $f(\boldsymbol{x}) < 0$. The predicted label, $\hat{y}$, for the point denoted with the red circle will then be $\hat{y} = \text{sgn}\{f(\boldsymbol{x})\} > 0$, and it is classified in class $+1$, to which the star points belong.

Our discussion to present the classification task was based on features that take numeric values. Classification tasks where the features are of categorical type do exist and are of major importance, too.

Regression

Regression shares to a large extent the feature generation/selection stage, as described before; however, now the output variable, y, is *not* discrete, but it takes values in an interval in the real axis or in a region in the complex numbers' plane. Generalizations to vector-valued outputs are also possible. Our focus here is on real variables. The regression task is basically a function (curve/surface) fitting problem.

We are given a set of training samples, $(y_n,\ \boldsymbol{x}_n)$, $y_n \in \mathbb{R}$, $\boldsymbol{x}_n \in \mathbb{R}^l$, $n = 1, 2, \ldots, N$, and the task is to estimate a function f, whose graph fits the data. Once we have found such a function, when a new sample $\boldsymbol{x}$, outside the training set, arrives, we can predict its output value. This is shown in Fig. 1.2. The training data in this case are the gray points. Once the function fitting task has been completed, given a new point x (red), we are ready to predict its output value as $\hat{y} = f(\boldsymbol{x})$. In the simple case of the figure, the function f is linear and thus its graph is a straight line.

The regression task is a generic one that embraces a number of problems. For example, in financial applications, one can predict tomorrow's stock market prices given current market conditions and other related information. Each piece of information is a measured value of a corresponding feature. Signal and image restoration come under this common umbrella of tasks. Signal and image denoising can also be seen as a special type of the regression task. Deblurring of a blurred image can also be treated as regression (see Chapter 4).

1.6 Unsupervised, self-supervised, and semi-supervised learning

The goal of supervised learning is to establish a functional relationship between the input and output variables. To this end, labeled data are used, which comprise the set of output–input pairs, on which the learning of the unknown mapping is performed.

In the antipode of supervised learning lies *unsupervised learning*, where only input variables are provided. No output or label information is available. The aim of unsupervised learning is to unravel the structure that underlies the given set of data. This is an important part in data learning methods. Unsupervised learning comes under a number of facets.

One of the most important types of unsupervised learning is that of *clustering*. The goal of any clustering task is to unravel the way in which the points in a data set are grouped, assuming that such a group structure exists. As an example, given a set of newspaper articles, one may want to group them together according to how similar their content is. As a matter of fact, at the heart of any clustering algorithm lies the concept of similarity, since patterns that belong to the same group (cluster) are assumed to be more similar than patterns that belong to different clusters.

One of the most classical clustering schemes, the so-called k-means clustering, is presented and discussed in Section 12.6.1. However, clustering is not a main topic of this book, and the interested reader may look at more specialized references (e.g., [14,15]).

Another type of unsupervised learning is *dimensionality* reduction. The goal is also to reveal a particular structure of the data, which is of a different nature than that of the groupings. For example, although the data may be represented in a high-dimensional space, they may lie around a lower-dimensional subspace or a manifold. Such methods are very important in machine learning for compressed representations or computational reduction reasons. Dimensionality reduction methods are treated in detail in Chapter 20.

Probability distribution estimation can also be considered as a special case of unsupervised learning. Probabilistic modeling is treated extensively in Chapters 12, 13, 15, and 16.

Self-supervised learning (SSL) is another family of algorithms where no labeled data are available but, instead, target values are *constructed* from unlabeled examples. Thus, one could consider self-supervised learning as a category of learning methods that lies between supervised and unsupervised ones. Self-supervised methods have attracted a lot of interest and attention in the recent years as a way to exploit the vast amount of unlabeled data that are available in the web and reside in various data sets. The typical idea behind a self-supervised learning task is first to *pretrain* a network as an encoder, via unlabeled data. This phase is known as the *pretext* task. The encoder can then be used as a feature generator for different tasks, where only a limited number of labeled data exist. This phase is known as the *downstream* task and parameter fine-tuning is performed in a *supervised* way using the available labeled examples. Self-supervised learning is treated in Chapter 19.

Semi-supervised is a another family of methods that lie in between supervised and unsupervised learning. In semi-supervised learning, there are labeled data but not enough to get a good estimate of the output–input dependence. The existence of a number of unlabeled patterns can assist the task, since it can reveal additional structure of the input data that can be efficiently utilized. Semi-supervised learning is treated in, e.g., [14].

Finally, another type of learning, which is increasingly gaining in importance, is the so-called *reinforcement learning* (RL). This is also an old field, with origins in automatic control. At the heart of this type of learning lies a set of rules and the goal is to learn sequences of actions that will lead an *agent*

to achieve its goal or to maximize its objective function. For example, if the agent is a robot, the goal may be to move from point A to point B. Intuitively, RL attempts to learn actions by *trial and error*. In contrast to supervised learning, optimal actions are not learned from labels but from what is known as a *reward*. This scalar value informs the system whether the outcome of whatever it did was right or wrong. Taking actions that maximize the reward is the goal of RL. Reinforcement learning is beyond the scope of this book and the interested reader may consult, e.g., [19].

1.7 Structure and a road map of the book

In the discussion above, we saw that seemingly different applications, e.g., authorship identification and channel equalization, as well as financial prediction and image deblurring, can be treated in a unified framework. Many of the techniques that have been developed for machine learning are no different than techniques used in statistical signal processing or adaptive signal processing. Filtering comes under the general framework of regression (Chapter 4), and "adaptive filtering" is the same as "online learning" in machine learning. As a matter of fact, as will be explained in more detail, this book can serve the needs of more than one advanced graduate or postgraduate courses.

Over the years, a large number of techniques have been developed, in the context of different applications. Most of these techniques belong to either of two schools of thought. In one of them, the involved parameters that define an unknown function, for example, $\theta_1, \theta_2, \theta_0$ in Eq. (1.1), are treated as random variables. Bayesian learning builds upon this rationale. Bayesian methods learn distributions that describe the randomness of the involved parameters/variables. According to the other school, parameters are treated as nonrandom variables. They correspond to a fixed, yet unknown value. We will refer to such parameters as deterministic. This term is justified by the fact that, in contrast to random variables, if the value of a nonrandom variable is known, then its value can be "predicted" exactly. Learning methods that build around deterministic variables focus on the choice of an optimizing loss function to obtain estimates of the corresponding values. In some cases, the term "frequentist" is used to describe the latter type of techniques (see Chapter 12).

Each of the two previous schools of thought has its pros and cons, and I firmly believe that there is always more than one road that leads to the "truth." Each can solve some problems more efficiently than the other. Maybe in a few years, the scene will be more clear and more definite conclusions can be drawn. Or it may turn out, as in life, that the "truth" is somewhere in the middle.

In any case, every newcomer to the field has to learn the basics and the classics. That is why, in this book, all major directions and methods will be discussed, in an equally balanced manner, to the greatest extent possible. Of course, the author, being human, could not avoid giving some more emphasis to the techniques with which he is most familiar through his own research. This is healthy, since writing a book is a means of sharing the author's expertise and point of view with the readers. This is why I strongly believe that a new book does not serve to replace previous ones, but to complement previously published points of view.

Chapter 2 is an introduction to probability and statistics. Random processes are also discussed. Readers who are familiar with such concepts can bypass this chapter. On the other hand, one can focus on different parts of this chapter. Readers who would like to focus on statistical signal processing/adaptive processing can focus more on the random processes part. Those who would like to follow a probabilistic machine learning point of view would find the part presenting the various distributions

more important. In any case, the multivariate normal (Gaussian) distribution is a must for those who are not yet familiar with it.

Chapter 3 is an overview of the parameter estimation task. This is a chapter that presents an overview of the book and defines the main concepts that run across its pages. This chapter has also been written to stand alone as an introduction to machine learning. Although it is my feeling that all of it should be read and taught, depending on the focus of the course and taking into account the omnipresent time limitations, one can focus more on the parts of her or his interest. Least-squares and ridge regression are discussed alongside the maximum likelihood method and the presentation of the basic notion of the Bayesian approach. In any case, the parts dealing with the definition of the inverse problems, the bias–variance tradeoff, and the concepts of generalization and regularization are a must.

Chapter 4 is dedicated to the mean-square error (MSE) linear estimation. For those following a statistical signal processing course, the whole chapter is important. The rest of the readers can bypass the parts related to complex-valued processing and also the part dealing with computational complexity issues, since this is only of importance if the input data are random processes. Bypassing this part will not affect reading later parts of the chapter that deal with the MSE estimation of linear models, the Gauss–Markov theorem, and the Kalman filtering.

Chapter 5 introduces the stochastic gradient descent family of algorithms. The first part, dealing with the stochastic approximation method, is a must for every reader. The rest of the chapter, which deals with the least-mean-squares (LMS) algorithm and its offsprings, is more appropriate for readers who are interested in a statistical signal processing course, since these families are suited for tracking time-varying environments. This may not be the first priority for readers who are interested in classification and machine learning tasks with data whose statistical properties are not time-varying.

Chapter 6 is dedicated to the least-squares (LS) method, which is of interest to all readers in machine learning and signal processing. The latter part, dealing with the total least-squares method, can be bypassed in a first reading. Emphasis is also put on ridge regression and its geometric interpretation. Ridge regression is important to the newcomer, since he/she becomes familiar with the concept of regularization; this is an important aspect in any machine learning task, tied directly with the generalization performance of the designed predictor.

I have decided to compress the part dealing with fast LS algorithms, which are appropriate when the input is a random process/signal that imposes a special structure on the involved covariance matrices, into a discussion section. It is the author's feeling that this is of no greater interest than it was a decade or two ago. Also, the main idea, that of a highly structured covariance matrix that lies behind the fast algorithms, is discussed in some detail in Chapter 4, in the context of Levinson's algorithm and its lattice and lattice-ladder by-products.

Chapter 7 is a must for any machine learning course. Important classical concepts, including classification in the context of the Bayesian decision theory, nearest neighbor classifiers, logistic regression, Fisher's discriminant analysis, and decision trees are discussed. Courses on statistical signal processing can also accommodate the first part of the chapter dealing with the classical Bayesian decision theory.

The aforementioned six chapters comprise the part of the book that deals with more or less classical topics. The rest of the chapters deal with more recent techniques and can fit with any course dealing with machine learning or statistical signal processing, depending on the focus, the time constraints, and the background of the audience.

Chapter 8 deals with convexity, a topic that is receiving more and more attention recently. The chapter presents the basic definitions concerning convex sets and functions and the notion of projec-

tion. These are important tools used in a number of recently developed algorithms. Also, the classical projections onto convex sets (POCS) algorithm and the set-theoretic approach to online learning are discussed as an alternative to gradient descent-based schemes. Then, the task of optimization of nonsmooth convex loss functions is introduced, and the family of proximal mapping, alternating direction method of multipliers (ADMM), and forward backward-splitting methods are presented. This is a chapter that can be used when the emphasis of the course is on optimization. Employing nonsmooth loss functions and/or nonsmooth regularization terms, in place of the squared error and its ridge regression relative, is a trend of high research and practical interest.

Chapters 9 and 10 deal with sparse modeling. The first of the two chapters introduces the main concepts and ideas and the second deals with algorithms for batch as well for as online learning scenarios. Also, in the second chapter, a case study in the context of time-frequency analysis is discussed. Depending on time constraints, the main concepts behind sparse modeling and compressed sensing can be taught in a related course. These two chapters can also be used as a specialized course on sparsity on a postgraduate level.

Chapter 11 deals with learning in reproducing kernel Hilbert spaces and nonlinear techniques. The first part of the chapter is a must for any course with an emphasis on machine learning. Support vector regression and support vector machines are treated in detail. Moreover, a course on statistical signal processing with an emphasis on nonlinear modeling can also include material and concepts from this chapter. A case study dealing with authorship identification is discussed at the end of this chapter.

Chapters 12 and 13 deal with Bayesian learning. Thus, both chapters can be the backbone of a course on machine learning and statistical signal processing that intends to emphasize Bayesian methods. The former of the chapters deals with the basic principles and it is an introduction to the expectation-maximization (EM) algorithm. The use of this celebrated algorithm is demonstrated in the context of two classical applications: linear regression and Gaussian mixture modeling for probability density function estimation. The second chapter deals with approximate inference techniques, and one can use parts of it, depending on the time constraints and the background of the audience. Sparse Bayesian learning and the relevance vector machine (RVM) framework are introduced. At the end of this chapter, nonparametric Bayesian techniques such as the Chinese restaurant process (CRP), the Indian buffet process (IBP), and Gaussian processes are discussed. Finally, a case study concerning hyperspectral image unmixing is presented. Both chapters, in their full length, can be used as a specialized course on Bayesian learning.

Chapters 14 and 17 deal with Monte Carlo sampling methods. The latter chapter deals with particle filtering. Both chapters, together with the two previous ones that deal with Bayesian learning, can be combined in a course whose emphasis is on statistical methods of machine learning/statistical signal processing.

Chapters 15 and 16 deal with probabilistic graphical models. The former chapter introduces the main concepts and definitions, and at the end it introduces the message passing algorithm for chains and trees. This chapter is a must for any course whose emphasis is on probabilistic graphical models. The latter of the two chapters deals with message passing algorithms on junction trees and then with approximate inference techniques. Dynamic graphical models and hidden Markov models (HMMs) are introduced. The Baum–Welch and Viterbi schemes are derived as special cases of message passaging algorithms by treating the HMM as a special instance of a junction tree.

Chapters 18 and 19 deal with neural networks and deep learning. Chapter 18 is also a must in any course on machine learning. The chapter starts from the early days of the perceptron algorithm

and perceptron rule and moves on to the most recent advances in deep learning. The feed-forward multilayer architecture is introduced and a number of stochastic gradient-type algorithmic variants, for training networks, are presented. Different types of nonlinearities and loss functions are presented and their interplay with respect to the vanishing/exploding phenomenon in the gradient propagation is discussed. Regularization and the dropout method are considered. Convolutional neural networks (CNNs) and recurrent neural networks (RNNs) are reviewed in some detail. The notions of attention mechanism and adversarial examples are presented. The chapter concludes with an introduction to capsule networks, and a brief discussion on transfer learning and some open problems.

Chapter 19 comprises the main addition of the third edition. It covers some of the more recent advances in deep learning. Generative models are considered, beginning with the family of deep belief networks and in the sequel generative adversarial networks (GANs), variational autoencoders and, finally, diffusion models are discussed. The application of text-to-image generation is outlined. Normalizing flows are briefly highlighted. Another major focus of the chapter is on transformers and large language models. The families of GPT and BERT are presented. Self-supervised learning is also treated in some detail, providing the main concepts as well as a number of algorithmic families, including masked prediction methods and contrastive techniques. Some ethical issues that are raised when using generative and large language models are also discussed.

Chapter 20 is on dimensionality reduction techniques and latent variable modeling. The methods of principal component analysis (PCA), canonical correlations analysis (CCA), and independent component analysis (ICA) are introduced. The probabilistic approach to latent variable modeling is discussed, and the probabilistic PCA (PPCA) is presented. Then, the focus turns to dictionary learning and to robust PCA. Nonlinear dimensionality reduction techniques such as kernel PCA are discussed, along with classical manifold learning methods: local linear embedding (LLE) and isometric mapping (ISOMAP). Finally, a case study in the context of functional magnetic resonance imaging (fMRI) data analysis, based on ICA, is presented.

Each chapter starts with the basics and moves on to cover more recent advances in the corresponding topic. This is also true for the whole book, and the first six chapters cover more classical material.

In summary, we provide the following suggestions for different courses, depending on the emphasis that the instructor wants to place on various topics.

- Machine learning:
  - Main chapters: 3, 7, 11, and 18. These days, some parts of Chapter 19 can also be considered.
  - Secondary chapters: 12 and 13, and possibly the first part of 6.
- Statistical learning and signal processing:
  - Main chapters: 3, 4, 6, and 12.
  - Secondary chapters: 5 (first part) and 13–17.
- Machine learning with emphasis on Bayesian techniques:
  - Main chapters: 3 and 12–14.
  - Secondary chapters: 7, 15, and 16, and possibly the first part of 6.
- Adaptive signal processing:
  - Main chapters: 3–6.
  - Secondary chapters: 8, 9, 10, 11, 14, and 17.

I believe that the above suggestions of following various combinations of chapters are possible, since the book has been written in such a way as to make individual chapters as self-contained as possible.

At the end of most of the chapters, there are computer exercises, mainly based on the various examples given in the text. The exercises in the book are given in MATLAB® and the respective code is available on the book's website. Moreover, all exercises are provided, together with respective codes, in Python and are also available on the book's website. Some of the exercises in Chapters 18 and 19 are in the context of Pytorch.

The solutions manual as well as all the figures of the book are available on the book's website. Online lectures for some of the chapters are also available, as it is indicated at the end of the Preface.

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